Quiz 5

Name: _____

AI 501: Mathematics for AI | Duration: 20 min | Student ID: _____
Total: 10 marks

Question 1 (2 marks)

For each statement place a tick to indicate whether it is **True** or **False**.

Statement	True	False
(i) For an $m \times n$ matrix A, if m is strictly greater than n, the column space $\mathcal{C}(A)$ can span all of $\mathbb{R}^m$.	☐	☐
(ii) For an $m \times n$ matrix A, If the system $Ax = 0$ has only the trivial solution $x = 0$, then the rank of A must be exactly n.	☐	☐

Solution:

Statement	True	False
(i) $m > n$ matrix column space spanning $\mathbb{R}^m$		✓
(ii) $Ax = 0$ trivial solution implies rank is n	✓	
(iii) Rank of AB can be strictly greater than rank of A		✓

Justifications:
(i) False. The column space is spanned by at most n independent vectors. If $m > n$, the dimension of the column space (rank) is at most n, so it cannot possibly span an m-dimensional space ($\mathbb{R}^m$).
(ii) True. If $\mathcal{N}(A) = \{0\}$, the nullity is 0. By the rank-nullity theorem ($n = \text{rank}(A) + \text{nullity}(A)$), the rank must be exactly n (it has full column rank).
(iii) False. The columns of AB are linear combinations of the columns of A. Therefore, the column space of AB is a subspace of the column space of A, meaning $\text{rank}(AB) \leq \text{rank}(A)$.
1 mark per correct part (0.5 for tick, 0.5 for justification).

Question 2 (4 marks)

You are building a 1-NN classifier to predict if a patient is **Low Risk** or **High Risk**. The dataset uses three features: Age, White Blood Cell count, and number of previous visits.

Training Data: P_1: (20, 5000, 0) → Low Risk. P_2: (65, 9500, 4) → High Risk.
Test Patient: T: (25, 9500, 1).

(a) Calculate the squared Euclidean distance from T to P_1 and P_2. What is the prediction?

(b) Larger numeric ranges can skew distance calculations. Apply Min-Max scaling to the features using the formula $x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$. Write down the new scaled coordinates for P_1, P_2, and T and recompute the L_2^2 distances. What is the prediction now?

(c) If you expand this dataset to 1,000 patients use 50-NN instead of 1-NN, does the **variance** increase or decrease? Does the **bias** increase or decrease? Briefly explain why.

Solution:

(a) $L_2^2(T, P_1) = (25-20)^2 + (9500-5000)^2 + (1-0)^2 = 25 + 20,250,000 + 1 = \mathbf{20,250,026}$.
$L_2^2(T, P_2) = (25-65)^2 + (9500-9500)^2 + (1-4)^2 = 1600 + 0 + 9 = \mathbf{1609}$.
Nearest Neighbor: P_2. **Prediction: High Risk**.

(b) Min/Max ranges: Age [20, 65, range 45], WBC [5000, 9500, range 4500], Visits [0, 4, range 4].
Scaled: $P_1 = (0, 0, 0)$; $P_2 = (1, 1, 1)$; $T = (5/45, 4500/4500, 1/4) = (\mathbf{0.11}, \mathbf{1}, \mathbf{0.25})$.
$L_2^2(T_{scaled}, P_{1_scaled}) = (0.11-0)^2 + (1-0)^2 + (0.25-0)^2 = 0.0121 + 1 + 0.0625 = \mathbf{1.0746}$.
$L_2^2(T_{scaled}, P_{2_scaled}) = (0.11-1)^2 + (1-1)^2 + (0.25-1)^2 = 0.7921 + 0 + 0.5625 = \mathbf{1.3546}$.
New Nearest Neighbor: P_1. **New Prediction: Low Risk**.

(c) Increasing k from 1 to 50 averages the predictions over more neighbors, which **decreases variance** (the boundary becomes smoother) and **increases bias** (the model makes stronger assumptions, losing the ability to capture highly complex local patterns).

Question 3 (4 marks)

A binary classifier is designed to detect a disease. Your test dataset contains exactly 10,000 patients, but the disease prevalence is only 1% (i.e., only 1% actually have the disease). At a chosen decision threshold t, the model achieves a Recall of 100%, but a Precision of 10%.

(a) Using the provided metrics and dataset size, reconstruct the full 2×2 Confusion Matrix (True Positives, False Positives, True Negatives, and False Negatives). Show your work.

(b) Calculate the overall Accuracy and the False Positive Rate (FPR) of the model.

Solution:

(a) Total Positives = 1% of 10,000 = 100. Total Negatives = 9,900.
Recall = 100% $\rightarrow$ Model found all 100 sick people. **TP = 100, FN = 0**.
Precision = $\frac{TP}{TP+FP} = 0.10 \rightarrow \frac{100}{100+FP} = 0.10 \rightarrow FP = 900$. **FP = 900**.
Since Total Negatives = 9,900, $TN = 9,900 - 900 = 9,000$. **TN = 9,000**.

(b) Accuracy = $\frac{TP+TN}{\text{Total}} = \frac{9,100}{10,000} = \mathbf{91\%}$.
FPR = $\frac{FP}{FP+TN} = \frac{900}{9,900} = \frac{1}{11} \approx \mathbf{9.09\%}$.

(c) The flaw is ignoring the sheer volume of False Positives (900 healthy people subjected to a false alarm). **No, an ROC curve would not expose this weakness.** ROC plots TPR vs. FPR. Because of the extreme class imbalance, the massive True Negative count (9,000) keeps the FPR artificially low ($\approx 9\%$).